

Altera FPGA

Intel HyperFlex FPGA Architecture and Intel's 14nm tri-gate process technology enable Intel Stratix 10 FPGAs and SoCs to deliver levels of performance and power efficiency that were unimaginable in previous-generation high-performance FPGAs. These devices deliver two times the core performance and five times the density compared to previous-generation Stratix V FPGAs.

In addition, these offer up to 70 per cent lower power than Stratix V FPGAs for equivalent performance, logic, internal memory and DSP blocks capable of 1GHz operation, embedded quad-core 64-bit ARM Cortex-A53 hard processor system (in SoC variants) and familiar FPGA design techniques supported by Intel Quartus Prime software.

The centrepiece of Intel HyperFlex FPGA architecture is its registers design that adds bypassable hyper-

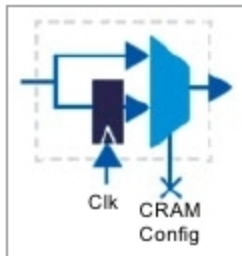


Fig. 5: Bypassable hyper-register

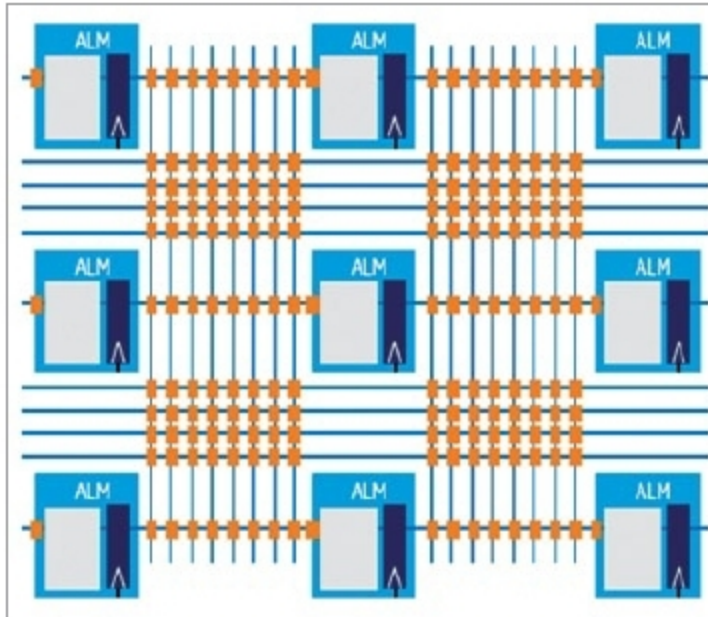


Fig. 6: Hyperflex architecture registers

registers to every routing segment in the FPGA core and at all functional block inputs. Fig. 5 shows a bypassable hyper-register, where the routing signal can bypass the register and go straight to the multiplexer, or go through the register first. The multiplexer is controlled by one bit of the FPGA configuration memory (CRAM).

Stratix 10 FPGA and SoC system integration includes heterogeneous 3D

system-in-package integration, highest memory bandwidth with package, dual-mode transceiver with 56Gbps PAM-4 and 30Gbps NRZ, dual-mode transceiver with 56Gbps PAM-4 and 30Gbps NRZ. It has the highest density FPGA fabric with up to 5.5 million LEs, up to 10 tera floating point operations per second of IEEE 754-compliant single precision floating-point digital signal processing throughput, secure device manager with the most comprehensive security capabilities, integrated quad-core 64-bit ARM

Cortex-A53 hard processor system up to 1.5GHz, and optimised and validated Intel Enpirion power solutions.

Stratix 10 devices are uniquely positioned to address the design challenges in next-generation, high-performance systems in virtually all end markets, including wireline and wireless communications, computing, storage, military, broadcast, medical, and test and measurement. **EFY**



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